

Math 111: Calculus Review for Exam 1

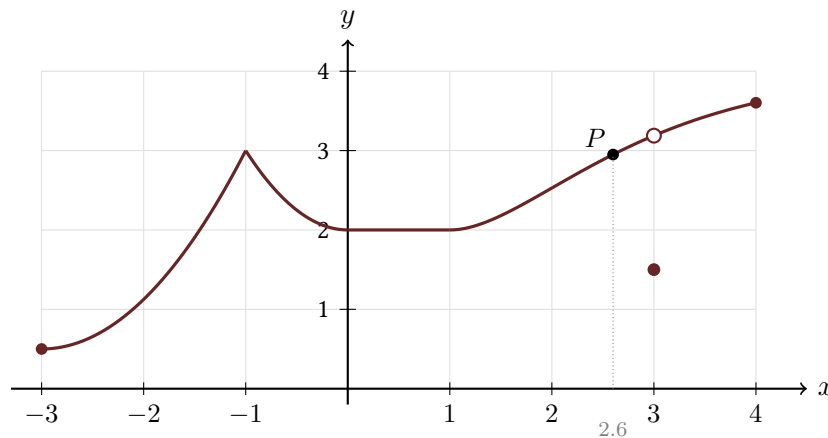
Wednesday, September 23 · Exam 1 is Friday, September 25, in class

Today is synthesis, not drill. Chapter 1 is not a list of separate skills. Almost every idea in it can be read off a single picture, and the exam rewards you for seeing that. We'll spend the hour putting the ideas next to one another.

Exam 1 in one paragraph. Four problems, fifty minutes, closed book and device-free. It covers everything through the tangent line approximation (§1.1, §1.3–§1.8). Each problem gets a holistic mark from 0 to 5. A 3 means the right idea with a major mistake, and that is worth far more than a blank. No shortcut derivative rules are needed. Everything comes from a picture, a table, or the definition.

Part 1. One graph, six lenses

Here is the graph of a function f on $[-3, 4]$. There's no formula, and you won't need one. Your group gets *one* lens. Work on it at your whiteboard, then we'll walk around and look at all six.



- Lens 1.** Where is $f' > 0$? Where is $f' = 0$? Where is f' undefined? Sketch f' underneath the graph, on the same horizontal scale.
- Lens 2.** Where is $f'' > 0$? Where is $f'' < 0$? Where does the concavity change?
- Lens 3.** Where does f fail to be differentiable? At each such place, say *which* condition fails.
- Lens 4.** Draw the tangent line L at P . Near P , does L over- or under-estimate f ? Why?
- Lens 5.** Read the graph as a temperature: $f(t)$ is the temperature in $^{\circ}\text{F}$, t hours after midnight. Write *one sentence each*, with units and an estimated number, about $f(1.5)$, $f'(1.5)$, and $f''(1.5)$.
- Lens 6.** Where does $\lim_{x \rightarrow c} f(x)$ exist even though f is not continuous at c ? Is there any c where the limit fails to exist? If not, describe a small change to the graph that would create one.

Part 2. Find the first wrong step

Here is a claim and an argument for it.

Claim. If f is continuous at a , then f is differentiable at a .

Argument. Since f is continuous at a , $\lim_{x \rightarrow a} f(x) = f(a)$, so $\lim_{x \rightarrow a} (f(x) - f(a)) = 0$. Also $\lim_{x \rightarrow a} (x - a) = 0$. Therefore

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \frac{0}{0} = 0,$$

which exists, so $f'(a) = 0$ exists.

(a) Which is the *first* line that is false? Say exactly what is wrong with it.

(b) Show that the claim is false, using a point on the graph in Part 1.

(c) The claim is true in the *other* direction. State it. *If you finish early:* prove it. Write

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a} \cdot (x - a)$$

and take $\lim_{x \rightarrow a}$ of both sides.

On Friday.

- Write what you know. A correct idea inside a wrong answer earns credit. A blank earns nothing.
- Label your diagrams: axes, the point, the line, which curve is which.
- A sentence of interpretation with units beats a bare number.
- Stuck on a part? Say what you would do, and move on.
- If the final goes better than your lowest in-class exam, it replaces that score.

Before Friday: the practice exam (forty minutes, book closed, phone in another room), Ezra's problem session Thursday 4:40–6:40, and office hours today 1:00–2:30 in Library 306.